Adam Acovski

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SUMMARY

Data scientist and data engineer with nearly a decade of predictive modeling experience, specializing in Python/SQL and scalable data pipelines for environmental and groundwater problems. Builds calibrated predictive models with rigorous uncertainty quantification (PEST++, iterative ensemble smoothers) and the production systems behind them: ETL pipelines, lakehouse storage, APIs, cloud infrastructure, and CI/CD. Domain depth from mining and environmental consulting, including groundwater flow and contaminant transport modeling, dewatering assessments, and model calibration.

TECHNICAL SKILLS

Modeling & Statistics: Bayesian inverse modeling, iterative ensemble smoothers (IES), uncertainty quantification, parameter estimation, scikit-learn, PyTorch
Data Engineering: ETL pipelines, data modeling, PostgreSQL, Cloud SQL, DuckDB, DuckLake, FastAPI, spatial-temporal data processing
Languages: Python (pandas, NumPy, SciPy, GeoPandas), SQL, Rust, C++, Git
Cloud & DevOps: GCP (Cloud Run, Cloud SQL, Cloud Storage, Compute Engine), AWS, Pulumi, Docker, GitHub Actions, Cloud Build, pytest
Visualization: Plotly, Dash, Matplotlib, React, Three.js, QGIS
Domain Tools: Leapfrog, MODFLOW 6, FloPy, FEFLOW, MINEDW, PEST++

EXPERIENCE

SRK Consulting – Consultant (Hydrogeology) *June 2026 – Present*

Lakewood, Colorado, United States

- Develop conceptual and numerical groundwater models that predict mine-site flow conditions for engineering decisions
- Work with exploration data in Leapfrog to inform conceptual hydrogeological models
- Design ETL pipelines on DuckDB and DuckLake that store monitoring, field, and model-output time series in normalized tables keyed by sensor and timestamp, with analysis-ready views for modeling and reporting
- Transform raw sensor data in-database, such as converting vibrating wire piezometer readings into water levels, so derived values stay traceable to source readings
- Serve analytical results to project teams through FastAPI services and React front ends

ITASCA Consulting – Project Hydrogeologist *July 2024 – May 2026*

Lakewood, Colorado, United States

- Calibrated transient groundwater models with FloPy and PEST++, using iterative ensemble smoothers for Bayesian parameter estimation and predictive uncertainty quantification
- Ran high-throughput ensemble simulations on GCP Compute Engine spot fleets orchestrated with Managed Instance Groups, cutting ensemble run times from weeks to days
- Automated conversion of transient 3D mine plans from non-native formats into MODFLOW 6 and MINEDW inputs, reducing a weeks-long manual task to seconds
- Performed QA/QC on geochemistry and well data for errors, gaps, and inconsistencies before use in analysis and reporting, and delivered geochemistry visualizations, such as Piper plots, through project dashboards
- Automated modeling workflows in Python to reduce manual processing, improve reproducibility, and support faster decision-making on spatial-temporal datasets
- Designed scheduled ETL pipelines that ingest groundwater-level, pumping, and water-quality monitor-

ing data from on-prem and cloud sources, validate and normalize it into Cloud SQL, and give model calibration and client reporting a single source of truth
- Structured pipelines so new wells, sites, and monitoring periods could be added without rework, keeping calibration inputs and reports consistent as datasets grew
- Delivered curated pipeline outputs through React/Plotly dashboards on GCP Cloud Run with IAP authentication, supporting ∼20 concurrent users across multiple firms on a single project
- Containerized applications with Docker for deployment to VPN-accessible servers
- Provisioned infrastructure as code with Pulumi and Docker, with CI/CD through GitHub Actions and Cloud Build and documentation published to AWS S3; unit tests guarded modeling scripts against regressions during refactoring

SNC-Lavalin – Groundwater Modeller *November 2012 – April 2020*

Saskatoon, Saskatchewan, Canada

- Calibrated ∼20 numerical groundwater models with PEST, estimating parameters by fitting simulations to observed field data and validating model performance before use in forecasting
- Optimized brine-disposal well-field configurations for the Winnipeg–Deadwood formation, using analytical and numerical models to forecast injection pressures and keep them below the fracture gradient
- Developed C++ FEFLOW plug-ins for geological model construction and mesh generation, including constrained Delaunay triangulation and mesh refinement that preserved geological layer boundaries, cutting model setup time from weeks to days; shapefile import replaced spreadsheet workflows that took days with near-instant processing
- Forecast long-term contaminant migration for uranium mine decommissioning using particle-tracking simulations, informing remediation strategies
- Translated model forecasts into technical reports for Environmental Impact Assessments (EIAs)
- Built and maintained HPC workstations for simulation workloads and managed hardware and software procurement

PROJECTS

Equity Options Trading (Self-employed) *April 2016 – Present*

- Manage an options portfolio using theta-decay strategies, sizing positions and managing risk through volatility and Greeks analysis

Game Development (Freelance, Rust/Bevy) *November 2021 – Present*

- Build a roguelike CRPG on an Entity-Component-System architecture, with procedural terrain generation (Voronoi diagrams, Perlin noise) and NPC pathfinding
- Optimize system performance and scalability for large, dynamic game worlds
- Contribute features and core-system improvements to open-source projects, including a multiplayer game server

EDUCATION

University of Waterloo *2007 – 2012*

Bachelor of Applied Science (BASc), Environmental/Environmental Health Engineering

CERTIFICATIONS

MSHA Training (surface and underground metal/non-metal mining)